Pratyush Shukla

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ACADEMIC AND PROFESSIONAL EXPERIENCE

Oct 2025 - Present

FPGA/SoC Developer @ *GMV, Munich*

- Developing FPGA and SoC-based architectures, implementing high-speed interfaces and protocols for space-grade, radiation-tolerant systems; performing HW/SW co-design and timing closure for on-board avionics.
- Conducting verification, validation, and certification of FPGA subsystems for satellite payloads and spacecraft instruments, focusing on reliable HW acceleration, interface integrity, and mission-critical performance.

Nov 2024 -Jun 2025

Master's Thesis @ *IEW, University of Stuttgart*

- Designed custom RTL IP cores (UWB emulation, SPI, Rx-TX sync, memory management) to support high-precision wireless charging tests on Cyclone V, Zynq 7000, and Ultrascale+ FPGA MPSoCs.
- Developed firmware enabling real-time UWB & WLAN positioning & data transfer.
- Verified RTL and timing closures to ensure system reliability.
- Implemented and verified FPGA-based UWB/WLAN system; performed real-world tests on vehicles to assess LOS obstruction, EMI, and packet loss.

Jun 2024 - Dec 2024

Werkstudent @ *Q.ANT, Stuttgart* | 🔗Ref.

- Designed RTL IP for data pipelines, memory access & multi-device sync on UltraScale+ MPSoC.
- Verified RTL with Cocotb and Verilog testbenches; handled timing violations & reset, and multi-device synchronisation.
- Designed high-speed ADC/DAC RTL for memory & stream management.
- Implemented Data handling solutions using BRAM, DDRx, and DMA pipelines.

Oct 2023 - Apr 2024

Embedded Intern @ *MAHLE, Stuttgart* | 🔗 *Ref.*

- Wrote firmware for signal processing, memory management on EEPROMs, auto-calibration of wireless charging devices, and CAN links.
- Worked on automating EV test benches & report generation with Python.

Oct 2023 - Mar 2024

Research Project (FA)@ ILH, *University of Stuttgart*

- Built transceivers on Zynq UltraScale+ for 75 GHz E-band using AD9082 MxFE & JESD204B for a low earth orbit satellite payload.
- Designed high-bandwidth data pipelines with BRAM, DDR3 & AXI-DMA.
- Worked on mixed-signal integration and specification validation to ensure system performance.

Apr 2023- Aug 2023

Werkstudent @ *Arena2036/SPIE, Stuttgart* | 🔗 *Ref.*

- Designed and deployed an industrial sensor mesh with heterogeneous nodes; developed embedded firmware for data acquisition and node control using MQTT, EnOcean, I^2C, and UART.
- Implemented InfluxDB–Grafana pipelines for high-volume data aggregation, storage, and real-time visualization.

Apr 2023- Jun 2023 **Student Assistant** (HiWi) @ IIS, *University of Stuttgart* | 🔗 Details

Accelerated AI model inference on PYNQ-ZU and Z2 using Xilinx DPU overlays and optimized FFT and spectral-averaging workflows via partial reconfiguration on Zynq UltraScale+ MPSoC.

Jul 2022- Nov 2022 **Student Assistant** (HiWi) @ IAS, *University of Stuttgart*

Wrote bare-metal code and scripts using Python and Cypher for context modeling using knowledge graphs and Neo4j databases for a medication assistance system.

Jun 2019- Aug 2021 **Software Engineer** @ *Indian Institute of Science, Bangalore, India* | 🔗*Ref.*

- Designed and simulated high-speed wireless protocols (IEEE 802.11ad, 60 GHz Wi-Gig, 802.11ax) in MATLAB WLAN Toolbox to evaluate video streaming performance in aircraft cabins.
- Developed an adaptive channel-allocation algorithm to reduce packet and bit error rates in aircraft cabins.
- Built energy-aware data pipelines for heat-harvesting IoT sensor nodes, implementing adaptive sampling to maximize uptime (*Publication*:🔗).
- Engineered media/control pipelines(UDP, SDR, TCP/IP) for 28 GHz 5G video-casting.
- Implemented real-time anomaly detection and corrective-action algorithms in Python for aircraft environmental-control data.

SKILLS

Programming & HDLs: VHDL, Verilog, C, C++, Python, SystemVerilog, Cocotb, UVM

Tools: Vivado, Vitis, Xilinx ISE, Quartus Prime, MPLAB, Multisim, Keil, MATLAB, Git.

Technologies
Devices: Zynq UltraScale+, Zynq 7000, Cyclone V, PYNQ, µcontrollers and µprocessors(dsPIC, MSP430, AVR/ATmega, ESP32 families)
High-Speed Interfaces: *AMBA AXI, Altera Avalon, JESD204 B/C, LVDS*
Communication Protocols: UART, SPI, I2C, CAN, UWB, MQTT, BLE, WLAN
Memory Interfaces: BRAM, DDRx, DMA pipelines

Languages :
German: Beginner
English: Full Professional Proficiency
Hindi: Full Professional Proficiency

EDUCATION

Apr 2022 - Jun 2025 **MSc, Electrical Engineering,** University of Stuttgart, Germany

Thesis: FPGA-based Evaluation of UWB and WLAN for EV Wireless Charging Systems.

Aug 2015 - May 2019 **B.Tech, Electronics and Communication Engineering**, DSCE, India

Thesis: A Fault-Tolerant, Low-Power Hexapod Robot for Navigating Disaster Zones

Publications

June 2025 P. Shukla, "Ultra-Wideband Based Synchronization Method for Bidirectional Wireless Power Transfer Systems". IEEE Wireless Power Technology Conference and Expo.

September 2020 P. Shukla, "Judicious data management for sustaining an energy harvesting sensor node" Concurrency and Computation: Practice and Experience, Wiley.